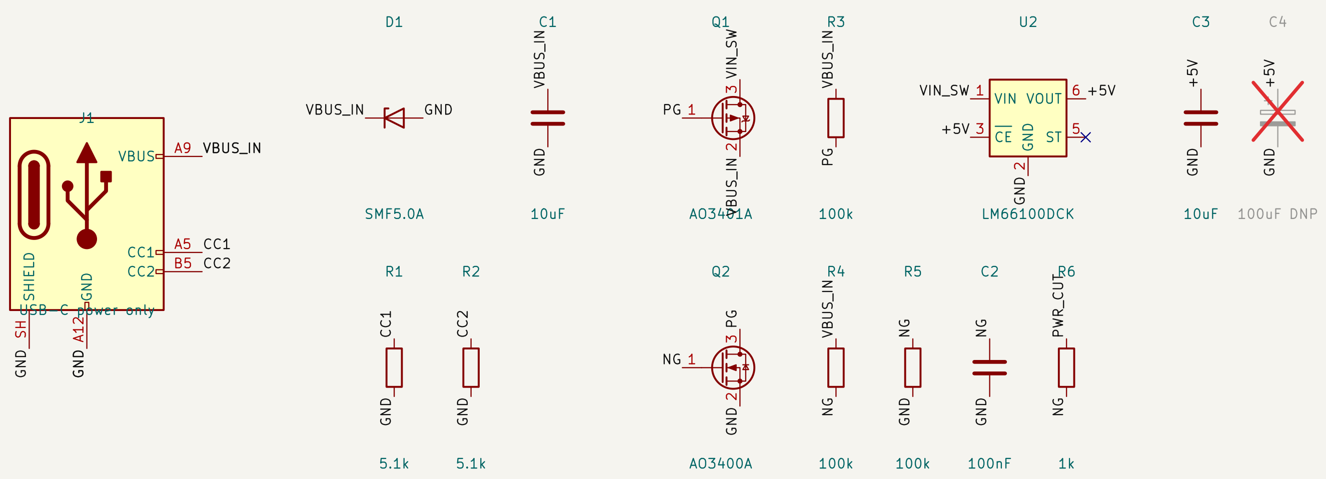
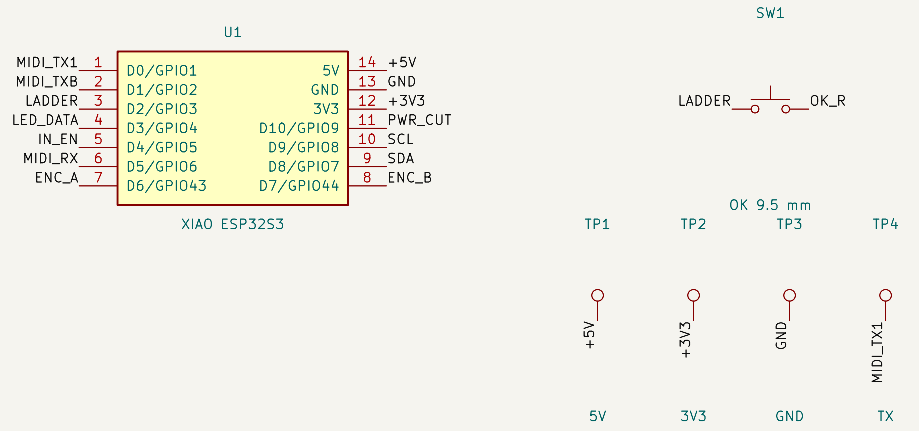
POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal−diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB−C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it. C4 (100 uF) not fitted: host−mode hot−plug reserve, against USB's 10 uF limit for a device.

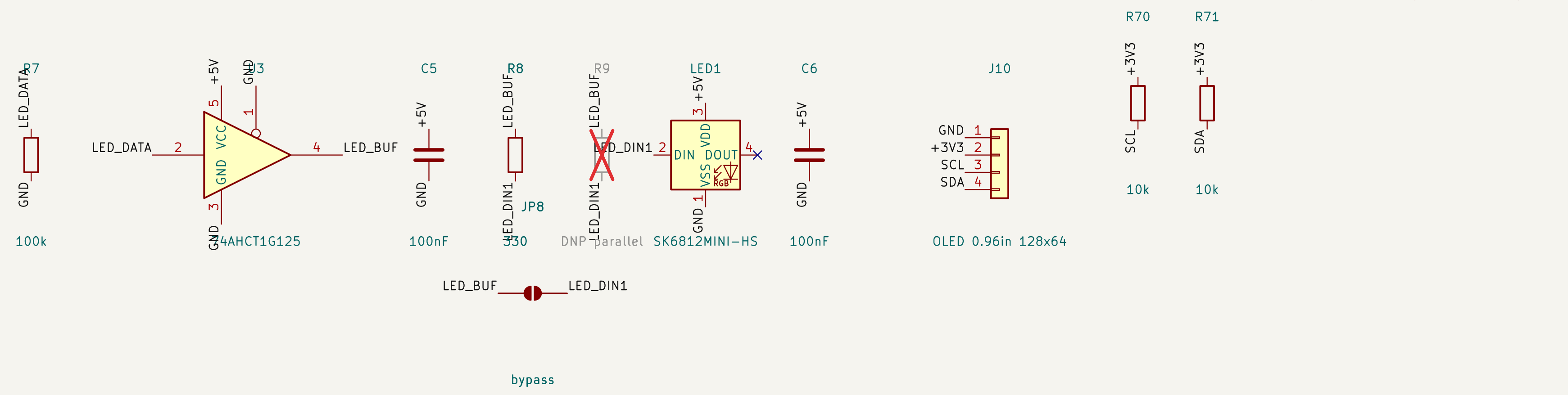


XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D4 = GPIO5, MIDI_RX = D5 = GPIO6 (v5).
v5: MIDI_TX1 = D0 = GPIO1 (OUT1 alone: Hi−Z in IN mode); MIDI_TXB = D1 = GPIO2 drives OUT2, OUT3 and OUT4 together (3 loops on one pad).
LADDER = D2 = GPIO3 (ADC1): OK / BACK / PUSH on one resistor ladder. LED_DATA = D3 = GPIO4. SCL = D8 = GPIO7, SDA = D9 = GPIO8 (the OLED, I2C by the GPIO matrix). PWR_CUT = D10 = GPIO9, low = J1 cut.



LED: one SK6812MINI−HS on +5V (top); U3 lifts the 3.3 V data to 5 V (VIH = 0.7 x VDD = 3.5 V at 5 V). SCREEN: J10, a 0.96 in 128 x 64 OLED (SSD1315, I2C 0x3C, 3V3)



On the board each pair is ONE footprint (J6..J9, jank:MIDI_OUT_PJ−320A_top_DIN−504B−M10_bottom: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; the DIN's pins on the bottom). J2..J5 are BOM lines only.

MIDI OUT 1−4 (CA−033 at 3.3 V), v5: BOTH fitted on every output, the TRS (type A, top) AND the DIN−5 (bottom), in parallel; plug one at a time

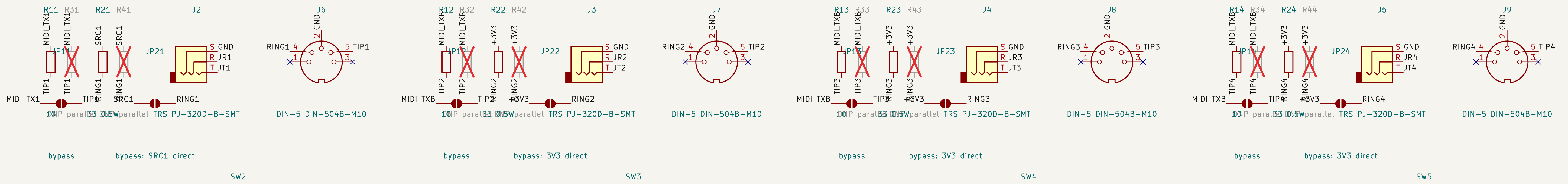
TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN−5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.

OUT1: J2 TRS and J6 DIN

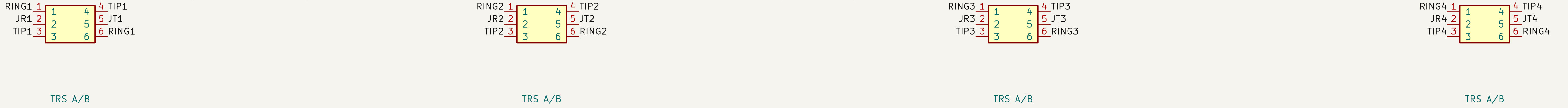
OUT2: J3 TRS and J7 DIN

OUT3: J4 TRS and J8 DIN

OUT4: J5 TRS and J9 DIN



Each MIDI resistor (and R8): axial, hand−swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.


RING1 JR1 TIP1 TIP1 JT1 RING1
RING2 JR2 TIP2 TIP2 JT2 RING2
RING3 JR3 TIP3 TIP3 JT3 RING3
RING4 JR4 TIP4 TIP4 JT4 RING4
TRS A/B

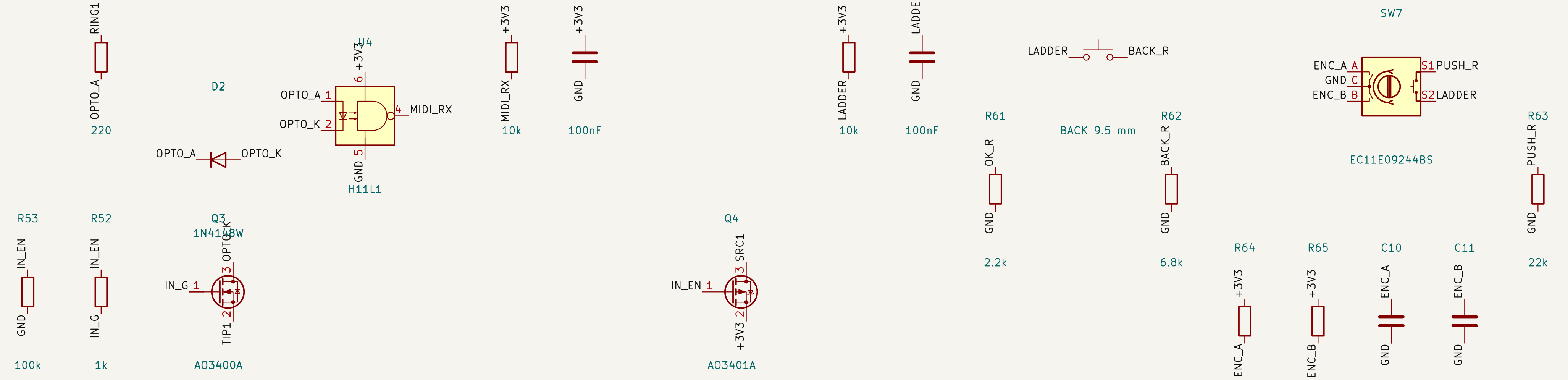
SW2..SW5: TRS type A / B per output (top side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1−2 / 4−5 closed = type A.

OUT1 AS MIDI IN (software): TX1 to input (Hi−Z), IN_EN high (Q4 off: no source), Q3 on), MIDI_RX by UART RX

CONTROLS (top side, by the screen): one ADC ladder on GPIO3: R60 10k to 3V3; OK 2.2k, BACK 6.8k, PUSH 22k to GND

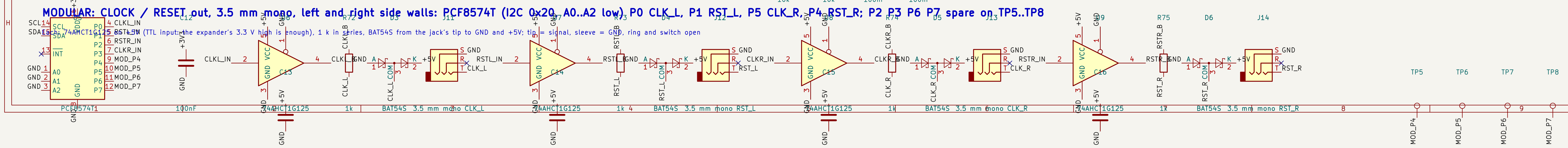
OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). OUT1 drives as before. D2 across the opto's LED (reverse protection). OUT1 sits on the BACK edge (long board).

Idle 3.30 V, OK 0.60 V, BACK 1.34 V, PUSH 2.27 V (one switch at a time). Encoder A/B: 10k to 3V3, 100 nF to GND each.



MODULAR: CLOCK / RESET out, 3.5 mm mono, left and right side walls: PCF8574T (I2C 0x20, A0..A2 low) P0 CLK_L, P1 RST_L, P5 CLK_R, P4 RST_R; P2 P3 P6 P7 spare on TP5..TP8

Each: 74AHCT1G125 on +5V (TTL input: the expander's 3.3 V high is enough), 1 k in series, BAT54S from the jack's tip to GND and +5V; tip = signal, sleeve = GND, ring and switch open



generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank−pcb−mod.kicad_sch

Title: jank: XIAO ESP32S3 MIDI clock stick

Size: A2
Date: 2026−11−08
Rev: 0.1 schematic for review
KiCad E.D.A. 10.0.8
Id: 1/1